

UNKNOWN QUANTITY

A REAL AND
IMAGINARY HISTORY
OF ALGEBRA

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ace at Sans Souci; Fourier was a scientific adviser on Napoleon's expedition to Egypt.

D'Alembert's pioneering work on differential equations in the middle of the century was characteristic, and Laplace's equation $\nabla^2\phi=0$, which describes numerous physical systems where a quantity (density, temperature, electric potential) is distributed smoothly but unequally across an area or a volume, can be taken as representative of applied math at the century's end.

Algebra was something of a bystander to all these glamorous developments. The general cubic and quartic equations had been cracked, but no one had much of a clue about how to proceed further in that direction. Viète, Newton, and one or two others among the most imaginative mathematicians had noticed the odd symmetries of the solutions of polynomial equations but had no idea how to make any mathematical profit from these observations.

There was, however, one other problem that mathematicians struggled with all through the 18th century and that I ought to cover here. This was the problem of finding a proof for the so-called fundamental theorem of algebra, hereinafter the FTA. I write "so-called" because the theorem always *is* so called, yet its status as implied by that name is considerably disputed. There are even mathematicians who will tell you, in the spirit of Voltaire's well-known quip about the Holy Roman Empire, that the FTA is neither fundamental, nor a theorem, nor properly within the scope of algebra. I hope to clarify all that in just a moment.

The FTA can be stated very simply, if a little roughly, in the context of polynomial equations, as: *Every equation has a solution*. To be more precise:

The Fundamental Theorem of Algebra

The polynomial equation $x^n + px^{n-1} + qx^{n-2} + \dots = 0$ in a single unknown x , the polynomial's coefficients $p, q, \dots$ being complex numbers, and n greater than zero, is satisfied by some complex number.

Ordinary real numbers are to be understood here as just particular cases of complex numbers, the real number a as the complex number $a + 0i$. So equations with real-number coefficients, like all the ones I have displayed so far, come under the scope of the FTA. Every such equation has a solution, though the solution may be a complex number, as in the case $x^2 + 1 = 0$, satisfied by the complex number i (and also by the complex number $-i$).

The FTA was first stated by Descartes in *La géométrie* (1637), though in a tentative form, as he was not at ease with complex numbers. All the great 18th-century mathematicians had a go at trying to prove it. Leibniz actually thought he had *disproved* it in 1702, but there was an error in his reasoning, pointed out by Euler 40 years later. The mighty Gauss made it the subject of his doctoral dissertation in 1799. Not until 1816 was a completely watertight proof given, though—also by Gauss.

To clarify the mathematical status of the FTA, you really need to study a proof. The proof is not difficult, once you have made friends with the complex plane (see Figure NP-4) and can be found in any good textbook of higher algebra.⁵⁹ What follows is the merest outline.

§6.5 *Proof of the Fundamental Theorem of Algebra*

It is the case with complex numbers, as with real numbers, that higher powers easily swamp lower ones, a thing I mentioned in §CQ.3. Cubes get seriously big much faster than squares, and fourth

powers much faster than cubes, and so on. (Note: The word “big,” when applied to complex numbers, means “far from the origin,” or equivalently “having a large modulus.”) For big values of x , therefore, the polynomial in that box above looks pretty much like x^n with some small adjustments caused by the other terms.

If x is zero, on the other hand, every term in the polynomial is equal to zero, except for the last, “constant,” term. So for tiny values of x , the polynomial just looks like that last constant term. (The constant term in $x^2 + 7x - 12$, for instance, would be -12 .)

If you change x smoothly and evenly, then x^2 , x^3 , x^4 , and all higher powers will also change smoothly and evenly, though at different speeds. They will not suddenly “jump” from one value to another.

Given those three facts, consider *all* the complex numbers x with some given large modulus M . These numbers, if you mark them in the complex plane, form the circumference of a perfect circle of radius M . The corresponding values of the polynomial form, but only approximately, the circumference of a much bigger circle, one with radius M^n . (If a complex number has modulus M , its square has modulus M^2 and so on. This is easy to prove.) That’s because x^n has swamped all the lower terms of the polynomial.

Gradually, smoothly, shrink M down to zero. Our perfect circle—all the complex numbers with modulus M —shrinks down to the origin. The corresponding values of the polynomial shrink down correspondingly, like a loop of rope tightening, from a vast near-circle centered on the origin to the single complex number that is the constant term in the polynomial. And in shrinking down like this, the tightening polynomial loop must at some point cross the origin. How else could all its points end up at that one complex number?

Which proves the theorem! The points of that dwindling loop are values of the polynomial, for some complex numbers x . If the loop crosses the origin, then the polynomial is zero, for some value of x . Q.E.D. (Though you might want to give a moment’s thought to the case where the constant term in the polynomial is zero.)

§6.6 The unhappy thing—from an algebraic point of view, I mean—about this proof is that it depends on the matter of *continuity*. I argued that as x changes gradually and slowly, so does the corresponding value of the polynomial. This is perfectly true, but it is only true because of the nature of the complex number system, in which you can glide without any jumps or bumps from one number to another, over the dense infinity of numbers in between.

Not all number systems are so accommodating. Number systems are many and various in modern algebra, and we can set up polynomials, and polynomial equations, in all of them. Not many are as friendly as the system of complex numbers, and the FTA is not true in all of them.

From the point of view of modern algebra, therefore, the FTA is a statement about a property of the complex number system, the property known in modern jargon as *algebraic closure*. The system of complex numbers (it says) is algebraically closed—which is to say, any single-unknown polynomial equation with coefficients in the system has a solution in the system. The FTA is not a statement about polynomials, equations, or number systems in general. That is why some mathematicians will take haughty pleasure in telling you that it is not fundamental; and while it is probably a theorem, it is not really a theorem in algebra but in analysis, where the notion of continuity most properly belongs.⁶⁰